

Diya Upadhyaya

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Portfolio: diyaupa.github.io

Education

University of South Florida

May 2027

B.S. in Computational and Applied Mathematics and Biomedical Sciences

Minor: Biomedical Physics

GPA: 3.92/4.0

Relevant Coursework: Linear Algebra, Graph Theory, Computational Methods for Applied Statistics, Optimization, Intermediate Analysis, Mathematical Modeling, Organic Evolution, Probability

Research Experience

MD Anderson Cancer Center

May 2026 – Present

Research Intern in PRiSM Lab — *PI: Dr. Heiko Enderling*

- Model tumor-immune dynamics in radiation and nanoparticle therapy-treated mice using a modified Lotka-Volterra system to evaluate treatment efficacy.
- Applied Latin hypercube sampling, genetic algorithms, and ODE solvers to assess parameter space and optimize treatment schedules.

North Carolina State University

May 2025 – Present

DRUMS REU Student — *PI: Drs. Mette Olufsen, Alen Alexanderian, Sujit Ghosh*

- One of twenty students selected nationwide for the Directed Undergraduate Research in Mathematics and Statistics (DRUMS) REU
- Develop a Valsalva Maneuver model utilizing a 7-ODE and 1-delay differential equation system.
- Implement uncertainty quantification techniques, including parameter subset selection and Hilbert-Schmidt Independence Criterion (HSIC), to reduce optimization problem size.
- Utilize MATLAB and FORTRAN to solve the ODE system and compute parameter identifiability and influence.

USF Department of Physics

Sept 2024 – Present

Researcher — *PI: Dr. Ghanim Ullah*

- Investigate the role of Piezo1 channels via Markov models in astrocytes through the dynamics of ions such as calcium, sodium, and ATP.
- Develop and solve the model using Euler's Method, programming primarily in Python and FORTRAN.

Integrated Mathematical Oncology, Moffitt Cancer Center

Sep 2023 – Present

Researcher and SPARK Intern — *PI: Dr. Renee Brady-Nicholls*

- Selected as one of sixteen students nationwide for a highly competitive research internship.
- Integrate patient-reported outcomes as prognostic biomarkers with ODE modeling to predict tumor volume dynamics in non-small cell lung cancer patients.
- Calibrate, validate, and generate predictive models primarily using MATLAB.
- **Workshop Researcher (Nov 2025):** Collaborated on patient-specific combinatorial immunotherapies in triple-negative breast cancer.
- **Workshop Researcher (Nov 2024):** Runner-up team; identified head and neck cancer patients at highest risk for treatment-induced toxicities.
- **Workshop Researcher (Oct 2023):** Winning team; developed a personalized AI decision tool for metastatic breast cancer.

Publications & Preprints

- Upadhyaya, D., Schabath, M., Hoogland, A., Brady-Nicholls, R. (2026). *Forecasting patient-specific tumor response using patient-reported outcomes in non-small cell lung cancer*. DOI: 10.64898/2026.01.29.26345069 (Preprint)

Presentations & Conferences

Poster Presentations

- Diya Upadhyaya, Renee Brady-Nicholls, "Patient Reported Outcomes as Indicators of Treatment Progression," OneUSF, Tampa, FL, April 2024.
- Diya Upadhyaya, Renee Brady-Nicholls, "Leveraging Mathematical Modeling with Patient-Reported Outcomes," Moffitt Cancer Center Scientific Symposium, Tampa, FL, May 2024.
- Diya Upadhyaya, Renee Brady-Nicholls, "Patient Reported Outcomes as Indicators of Treatment Progression," Oktoberfest Quantitative Sciences Division at Moffitt Cancer Center, Tampa, FL, October 2024.
- Diya Upadhyaya, Renee Brady-Nicholls, "Patient Reported Outcomes as Indicators of Treatment Progression," National Research Conference at University of Pennsylvania, Philadelphia, PA, October 2024.
- Diya Upadhyaya, Pawan Thapaliya, Ghanim Ullah, Ruhma Syeda, "Investigating the Role of PIEZO1 Channels in Astrocytic Function," The Biophysical Society Annual Meeting of 2025, Los Angeles, CA, Feb 2025.
- Camden Tumbleston, Diya Upadhyaya, Carter Hall, Troy Larsen, Sujit Ghosh, Alen Alexanderian, Mette S. Olufsen, "A Model of the Valsalva Maneuver for Autonomic Dysfunction," Summer Undergraduate Research & Creativity Symposium at North Carolina State University, Raleigh, NC, July 2025.
- Diya Upadhyaya, Renee Brady-Nicholls, "Incorporating The Tumor Growth Inhibition Model With Insomnia to Predict Time to Progression," MathOnco, St. Petersburg, FL, Oct 2025.
- Diya Upadhyaya, Renee Brady-Nicholls, "Forecasting patient-specific tumor response using patient-reported outcomes in non-small cell lung cancer," Joint Mathematics Meeting, Washington, D.C., Jan 2026.
- Camden Tumbleston, Diya Upadhyaya, Carter Hall, Troy Larsen, Sujit Ghosh, Alen Alexanderian, Mette S. Olufsen, "Preserving Physiological Fidelity in Mathematically Modeling the Valsalva Maneuver," Joint Mathematics Meeting, Washington, D.C., Jan 2026.

Oral Presentations

- Diya Upadhyaya, Renee Brady-Nicholls, "Patient-Reported Outcomes as Indicators of Treatment Progression in Non-Small Cell Lung Cancer," SPARK Intern Research Day, Tampa, FL, Aug 2024.
- Diya Upadhyaya, Ghanim Ullah, "Investigating the Role of the Mechanosensitive Ion Channel Piezo1 in the Pathophysiology of Astrocytes," Gulf Coast Undergraduate Research Conference, Houston, TX, Oct 2025.
- Diya Upadhyaya, Malgorzata Weh, Nahum-Puebla Osorio, Heiko Enderling "Modeling Tumor-Immune Dynamics Following Fractionated Radiation Therapy" Radiation Oncology Summer Student Symposium presentation, Houston, Texas, July 31, 2026.
- Diya Upadhyaya, Malgorzata Weh, Nahum-Puebla Osorio, Heiko Enderling "Modeling Tumor-Immune Dynamics Following Fractionated Radiation Therapy" Image Guided Cancer Therapy Research Program, Houston, Texas, August 3, 2026.

Mini-Symposiums

- Diya Upadhyaya, Renee Brady-Nicholls. "Forecasting Patient Specific Treatment Response by Leveraging a Mathematical Model and Patient-Reported Outcomes." Minisymposium presentation at the 14th European Conference on Mathematical and Theoretical Biology (ECMTB 2026), Graz, Austria, July 13–17, 2026. <https://ecmtb2026.org/contributions/ms74-04>.

Awards and Honors

- **2026 Goldwater Scholar**
- **Member of Sigma Xi**
- **2nd Place for Best Presentation:** Image Guided Cancer Therapy Symposium, MD Anderson
- **Finalist and Second Place Winner:** 3-Minute Thesis Competition, DRUMS REU
- **Full Travel Award Recipient:** Gulf Coast Undergraduate Research Conference (2025)
- **Honorable Mention:** Applied Physics Presentation, GCURS
- **Grant Recipient (\$50,000):** 2023 IMO Department Workshop Winning Team
- **Grant Recipient (\$50,000):** 2024 IMO Department Workshop Runner-up Team
- **Presidential Scholarship:** University of South Florida
- **Dean's List of Scholars:** Summer 2023 – Spring 2024, Spring 2025, Spring 2026

Skills

- **Technical Skills:** Python, MATLAB, R, Fortran, LaTeX
- **Languages:** English, Gujarati
- **Research Interests:** Mathematical Oncology, Computational Biology

Leadership and Involvement

Bioinformatics Club, USF

May 2024 – Present

President (May 2025 – August 2026) — Treasurer (May 2024 – April 2025)

- Manage a team of bioinformaticians and coordinate technical workshops on topics like AlphaFold and Kaggle to foster a collaborative educational environment.

Minorities in Medicine, USF

Jan 2024 – Aug 2025

Senior Finance Chair

- Oversaw club finances, organized healthcare panels, and launched a scholarship initiative raising over \$2,000 for pre-health students from low-socioeconomic backgrounds.

Teaching and Mentorship

- **USF Athletics Tutor** (*Aug 2025 – Dec 2025*): Tutored student-athletes in calculus, statistics, and physics.
- **Student Mentor** (*Dec 2024 – May 2025*): Empowered underrepresented students in medicine with personalized support and resources through Minorities in Medicine.
- **Calculus Peer Leader** (*Aug 2024 – Dec 2024*): Taught 50-student classes in Calculus and Engineering Calculus 1.
- **Chemistry Peer Leader** (*Jan 2024 – May 2024*): Facilitated General Chemistry 1 sessions for 10-20 students, fostering collaborative learning.

Community Involvement

- **Shadowing, Eichenfield Surgery Center** (*Mar 2024*): Monitored patient vitals and observed angioplasty and catheter removal procedures.
- **Volunteer, Moffitt Cancer Center** (*Sept 2023 – May 2024*): Supported patients in the CT scan department, ensuring a comfortable experience and escorting them to scanning rooms.